Critical Cities

Time limit: 1.00 s **Memory limit:** 512 MB

There are n cities and m flight connections between them. A city is called a *critical city* if it appears on every route from a city to another city.

Your task is to find all critical cities from Syrjälä to Lehmälä.

Input

The first input line has two integers n and m: the number of cities and flights. The cities are numbered $1, 2, \ldots, n$. City 1 is Syrjälä, and city n is Lehmälä.

Then, there are m lines describing the connections. Each line has two integers a and b: there is a flight from city a to city b. All flights are one-way.

You may assume that there is a route from Syrjälä to Lehmälä.

Output

First print an integer k: the number of critical cities. After this, print k integers: the critical cities in increasing order.

Constraints

- $2 \le n \le 10^5$
- $1 \le m \le 2 \cdot 10^5$
- $1 \le a, b \le n$

Example

Input	Output
5 5 1 2 2 3 2 4 3 5 4 5	3 1 2 5